

- 5 explicit 'substitution' in hypothesis wp.

(213)

int a=0;	<u>1. bsh</u> T		
read(Ra);	$\neg(a > 0)$		
if (a=0) {	T	$(\exists b_1: (\exists b_0: b_0 \neq 0 \wedge b_0 = b_1 - 1) \wedge a = 0)$ <u>$b_1 = a + 1$</u>	
b = a + 1;	T	$(\exists b_0: b_0 \neq 0 \wedge b_0 = b - 1) \wedge a = 0$	<u>3. bsh</u> T
if (a=0) {	$\neg(a > 0)$	$\exists b_0: b_0 \neq 0 \wedge b_0 = b - 1$	$b = a + 1$
b = b - 1;	$\neg(a > 0)$	$\neg(b_0 \neq 0 \wedge b_0 = b - 1)$	$\neg(a > 0) \wedge a = 0 \wedge b = a + 1$
if (b ≠ 0)	$\neg(a > 0)$	$T \wedge b \neq 0 = b \neq 0$	$b = 1 \wedge \neg(b = 0)$
assert;	$\neg(a > 0)$	<u>SAT? v</u>	$\neg(a > 0) \wedge a = 0 \wedge \neg(b = a + 1) \wedge \neg(b = 1)$
}		<u>2. bsh</u>	$\neg(b = 0)$
}			X unreachable

weak proof of $T \wedge (\exists b_1 (\exists b_0: b_0 \neq 0 \wedge b_0 = b_1 - 1) \wedge a = 0) \wedge b_1 = a + 1$

$\equiv \exists b_1 \exists b_0: b_0 \neq 0 \wedge b_0 = b_1 - 1 \wedge a = 0 \wedge b_1 = a + 1$ (pre cond dep)

$\equiv$ X

differs:

- $\neg(a = 0) \wedge \neg(b_1 = a + 1) \Rightarrow \neg(b_1 = 1)$
- $b_1 = 1 \wedge b_0 = b_1 - 1 \Rightarrow b_0 = 0$
- $b_0 \neq 0 \wedge b_0 = 0 \Rightarrow$ false

$\exists b_1 = \{a > 0, a = 0, b = a + 1\}$

$b = 1, b \neq b - 1, b = 0\}$

interleaf

varie predikate sledovane po
na lozila rastvaraj.

3/3

$\neg(a > b) \rightarrow$

T	T
$\neg(a > 0)$	T
T	T
$\neg(a > 0)$	T
$\neg(a > 0)$	T

interpretation

T	T
T	T
$b = a + 1$	T
$b = 1$	T
$b = 0$	T

was ist passiert? schlechtere Punkte
na höhere Reichweite.

$b = a + 1$

$\neg(a > 0) \wedge$ ~~$b = 1$~~

$\neg(a > 0) \wedge b = 0$

X

$$\frac{0}{0} = 0$$

$\Gamma \wedge \underline{\quad} \Rightarrow \varphi.$
 $a_1 = 0 \wedge$
 $b_1 = a_2 + 1 \wedge$
 $a_2 = 0 \wedge$
 $b_2 = b_1 - 1 \wedge$
 $b_2 \neq 0$

dann ist $I(b_1) \equiv b_1 = 1$
 dann ist $I(b_2) = b_2 = 0$

$\varphi \Rightarrow I$
 $\pm(b_1, a_2) \equiv (b_1 = a_2 + 1)$
 $\varphi \wedge I$ unzulässig

also female γ_{USAT} : $-b_1 = a_2 + 1 \wedge a_2 = 0 \Rightarrow b_1 = 1$

$$\begin{aligned} -b_2 &= b_1 - 1 \wedge b_1 = 1 \Rightarrow b_2 = 0 \\ -b_2 &= 0 \wedge b_2 \neq 0 \Rightarrow \text{false.} \end{aligned}$$